

AP CSP | Lesson 22: Efficiency & Project

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Complete all 12 original questions (30 marks, about 40 minutes). Q5 requires exactly TWO choices. Trace statements in their execution order. Each program statement and each brace occupies its own line. Show reasoning and use the stated valid input range.

1. For sessions $\leftarrow [20,30,45,10]$, what is sessions[3]? [1 mark]

A. 20 B. 30 C. 45 D. 10

Answer: _____

2. Which growth pattern is classified as unreasonable time in AP CSP? [1 mark]

A. Constant B. n C. n^2 D. 2^n

Answer: _____

3. Why might a heuristic be chosen for a time-limited optimization task? [1 mark]

A. It always proves the optimum. B. It may provide a useful result when guaranteed optimization is impractical.
C. It turns every undecidable problem into a decidable one. D. It guarantees zero computation.

Answer: _____

4. Which statement best describes an undecidable decision problem? [1 mark]

A. No algorithm can always terminate with the correct yes/no answer for every input instance. B. Its currently available computer is too slow.
C. Nobody has typed its program yet. D. Every single instance is impossible to solve.

Answer: _____

5. Which TWO statements are correct? Select exactly TWO. [1 mark]

A. Binary search requires sorted data for its elimination rule. B. CAN_MOVE(left) moves the robot left.
C. A procedure with parameters can reuse steps for different argument values. D. RETURN always displays its value directly to the user.

Answer: _____

6. countGoals counts list elements at least as large as goal. What does countGoals([20,30,45,10],30) return? [1 mark]

A. 1 B. 2 C. 4 D. 75

Answer: _____

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Written responses | Questions 7-8

7. [4 marks]

Let values $\leftarrow$ [10,20,30]. Then APPEND(values,40) executes. State values[1], LENGTH(values), the sequence of values visited by FOR EACH item IN values, and how many times its body executes.

8. [4 marks]

- (a) State the prerequisite for binary search.
- (b) For [3,8,12,19,27,34,50], give binary search's checked values for target 34.
- (c) State the effect of ROTATE_LEFT on position.
- (d) State what happens after an attempted blocked MOVE_FORWARD.

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Written responses | Questions 9-10

9.

[4 marks]

An algorithm performs n^2 core operations; another performs 2^n . For $n=8$, give both counts. Classify each as reasonable or unreasonable time in the AP sense. Explain why a heuristic might be considered when exact optimization is impractical.

10.

[4 marks]

Explain a decidable decision problem, an undecidable decision problem, why a slow exact algorithm does not by itself imply undecidability, and whether some instances of an undecidable problem can be solved.

Written responses | Questions 11-12

Write PROCEDURE countGoals(sessions,goal). Initialize hits to 0, traverse sessions, add 1 for each value at least goal, and RETURN the final count after the traversal. Use AP pseudocode, with each brace on its own line.

- Give count and feedback for [29,30,31] with goal 30.
- Give count and feedback for [0,0] with goal 30.
- Explain how the sessions list manages complexity.
- Explain how the goal parameter supports reuse.